

Semantic Manipulation — Evaluation Tables

xArm · VLA (π -style) · datasets: coffee 1×1 & 4×4, pouring 5×3, mug-tree 5×1 · trained at 50 / $\sim$ 100 / 150 demos (filled: coffee @50 & @150, pouring @105, mug-tree @150). Marks: ✓ success, ✗ failure; superscript = grasp (S/T) and note numbers.

Codes. Cups: WB White-Basic, O Orange, Bk Black, R Red (ID); Br Brown, WC White-Ceramic, Gr Gray, P Pink (OOD); TWC Tall-White-Ceramic. Machines: K Keurig, Bk Black, T Teal, W White (ID); R Red, TS Tastyle, Bl Blue (OOD). Bowls: BL Blue, LB Light-Blue, BK Black (ID); P Pink, TB Tall, W White (OOD).

Summary

Table 1: Success rate by condition (VLA). Coffee shown at 50 and 150 demos to expose dataset-size scaling.

Run	ID/ID	ID/OOD	OOD/ID	OOD/OOD	Overall
Coffee 4x4 VLA @150	75% _{12/16}	67% _{8/12}	62% _{10/16}	42% _{5/12}	62% _{35/56}
Coffee 4x4 VLA @50	44% _{7/16}	58% _{7/12}	12% _{2/16}	50% _{6/12}	39% _{22/56}
Coffee 1x1 VLA @150	0% _{0/16}	25% _{3/12}	6% _{1/16}	17% _{2/12}	11% _{6/56}
Coffee 1x1 VLA @50	6% _{1/16}	8% _{1/12}	19% _{3/16}	17% _{2/12}	12% _{7/56}
Pouring 5x3 VLA @105	83% _{10/12}	58% _{7/12}	50% _{6/12}	25% _{3/12}	54% _{26/48}

Training runs & dataset sizes

Task 1 — Cup on Coffee Machine (rows: machines, cols: cups)

Task 2 — Cup Pouring (rows: bowls, cols: cups)

Task 3 — Hang Cup on Mug Tree (5 trials per cup)

Appendix A — Coffee 4x4 VLA @150 demos (OLD run)

Footnote legends

Coffee notes (150-demo & OLD sheets). **1** Pushed machine back; **2** Tilted placement and contact on plate; **3** Took double time; **4** Pushed cup while trying to grasp; **5** Dropped cup on edge of platform; **6** Self collision; **7** Spilled cup; **8** Force-torque trigger; **9** Grasped inside cup; **10** Gripper hooked on handle; **11** Cannot lift mug; **O** Completed but took $\geq$ 1 min; ***** Not fully in cup holder; ****** Released while tilted

Table 2: Training runs and dataset sizes, cross-listed with *Model Train Baselines*. Each task is trained at 50 / ~ 100 / 150 demos; the eval-done rows match the grids in this report.

Dataset	Size	Train	Done	Eval	xArm
Brick_in_Drawer	150	Done	Tue	–	–
Brick_in_Drawer	50	Done	–	–	–
Brick_in_Drawer	100	Queued	–	–	–
Faucet	50	Queued	–	–	–
Mug Tree Task	150	Done	Tue	done	✓
Mug Tree Task	50	Done	Tue	–	–
Mug Tree Task	100	Done	–	–	–
Mug/Machine 1by1	50	Done	Tue	done	✓
Mug/Machine 1by1	100	Training	Tue	no	–
Mug/Machine 1by1	150	Done	–	done	✓
Mug/Machine 4by4	150	Done	Tue	done	✓
Mug/Machine 4by4	50	Done	–	done	✓
Mug/Machine 4by4	100	Done	Tue	no	✓
Pour Task	105	Done	Tue	done	✓
Pour Task	50	Done	Wed	no	✓
Pour Task	150	–	–	no	–

Coffee notes (50-demo sheets). **1** Spilled the cup; **2** Failed to grasp (hit cup, grasped air); **3** Hit the coffee machine; **4** Good pose, high Z; **5** Wrong grasp; **6** Torque sensor error; **7** Moved machine back; **8** Handle hooked on gripper; **9** Balancing on edge; **10** Dropped cup elsewhere

Pouring notes. **1** Bad return cup to table; **2** Went in for second pour; **3** A bit off-center for pour; **4** Tipped bowl while returning cup; **5** Started pushing cup down & didn't move; likely F/T error; **6** Tapped bowl while placing down; **7** Picked up cup and stopped moving; **8** Tall bowl was pushed while attempting to pour; **9** Two partial pours - i completed; **10** Gripper opened while pour attempt; **11** Severely off center; **12** Tipped cup over while picking up cup; **STS** Succeeded only after trying all three grasp attempts (side - i top - i side)

Mug-tree notes. **1** Pushed the tree; **2** Hung on edge, so dropped cup; **3** Dropped mug while not on branch; **4** Tilted grasp on mug lip; **5** Hanging mug on edge; **6** Arm errored out; **7** Tried to grab entire cup

Table 3: Coffee 4x4 VLA @150 demos — ID Cup / ID Machine (12/16, 75%).

Machine	WB	O	Bk	R
K	✓ ^S	✗ ^S	✓ ^S	✓ ^T
Bk	✓ ^S	✓ ^S	✗ ^T	✓ ^{S 1}
T	✓ ^{T 1}	✓ ^{T 3}	✓ ^{T 1}	✗ ^S
W	✓ ^{T 4}	✓ ^T	✓ ^{T 4}	✗ ^T

Table 4: Coffee 4x4 VLA @150 demos — ID Cup / OOD Machine (8/12, 67%).

Machine	WB	O	Bk	R
R	✓ ^S	✓ ^{T 1}	✓ ^{T 1,5}	✗ ^{S 6}
TS	✓ ^{S 4}	✗ ^S	✗ ^{S 7}	✗ ^S
Bl	✓ ^{S 1}	✓ ^{S 1}	✓ ^T	✓ ^{S 2}

Table 5: Coffee 4x4 VLA @150 demos — OOD Cup / ID Machine (10/16, 62%).

Machine	Br	WC	Gr	P
K	✓ ^{S 2}	✓ ^S	✓ ^S	✓ ^S
Bk	✗ ^{T 8,9}	✓ ^S	✓ ^S	✗ ^S
T	✓ ^{T 1,2,8}	✗ ^{S 10}	✗ ^{T 1,9}	✓ ^S
W	✓ ^{S 2}	✗ ^{T 4}	✓ ^{T 2}	✗ ^{S 6}

Table 6: Coffee 4x4 VLA @150 demos — OOD Cup / OOD Machine (5/12, 42%).

Machine	Br	WC	Gr	P
R	✗ ^S	✗ ^S	✗ ^T	✓ ^S
TS	✓ ^{S 1}	✗ ^S	✓ ^{S 2}	✗ ^S
Bl	✗ ^{S 1,7,8}	✗ ^{S 8}	✓ ^{S 1}	✓ ^S

Table 7: Coffee 1x1 VLA @150 demos — ID Cup / ID Machine (0/16, 0%).

Machine	WB	O	Bk	R
K	✗ ^T	✗ ^{T 11}	✗ ^T	✗ ^{T 11}
Bk	✗ ^{S 8}	✗ ^{T 11}	✗ ^T	✗ ^{T 11}
T	✗ ^{T 1,7}	✗ ^T	✗ ^T	✗ ^{T 7}
W	✗ ^{T 1,8}	✗ ^{T 11}	✗ ^T	✗ ^{S 11}

Table 8: Coffee 1x1 VLA @150 demos — ID Cup / OOD Machine (3/12, 25%).

Machine	WB	O	Bk	R
R	✓ ^{T 1}	✗ ^S	✗ ^{S 11}	✗ ^{S 7}
TS	✓ ^S	✗ ^S	✗ ^{T 11}	✗ ^{T 6}
Bl	✓ ^S	✗ ^{S 5}	✗ ^{T 11}	✗ ^{T 7}

Table 9: Coffee 1x1 VLA @150 demos — OOD Cup / ID Machine (1/16, 6%).

Machine	Br	WC	Gr	P
K	✗ ^{T 11}	✓ ^S	✗ ^T	✗ ^{S 7,8}
Bk	✗ ^S	✗ ^S	✗ ^T	✗ ^S
T	✗ ^S	✗ ^{T 7}	✗ ^{S 11}	✗ ^{S 7}
W	✗ ^T	✗ ^T	✗ ^T	✗ ^{S 10}

Table 10: Coffee 1x1 VLA @150 demos — OOD Cup / OOD Machine (2/12, 17%).

Machine	Br	WC	Gr	P
R	✗ ^{T 11}	✗ ^S	✗ ^{S 7}	✗ ^{T 7,9}
TS	✗ ^S	✓ ^S	✗ ^S	✗ ^T
Bl	✓ ^{S 3,5}	✗ ^{T 7}	✗ ^{S 11}	✗ ^{T 5,7}

Table 11: Coffee 1x1 VLA @150 demos — White-Basic re-run.

Machine	Result	Grasp	Notes
K	✓	S	redo: 1st attempt F
Bk	✗	S	–
T	✓	T	2
W	✗	T	1, 7, 10
R	✓	S	1
TS	✓	S	1
Bl	✗	T	7, 9

Table 12: Coffee 4x4 VLA @50 demos — ID Cup / ID Machine (7/16, 44%).

Machine	WB	O	Bk	R
K	✗ ^{S 2}	✗ ^{S 2}	✗ ^S	✓ ^S
Bk	✗ ^{S 2}	✗ ^{S 1}	✓ ^S	✗ ^S
T	✓ ^S	✓ ^S	✗ ^{S 1}	✗ ^{S 1}
W	✓ ^S	✓ ^S	✓ ^T	✗ ^{T 2}

Table 13: Coffee 4x4 VLA @50 demos — ID Cup / OOD Machine (7/12, 58%).

Machine	WB	O	Bk	R
R	✓ ^S	✓ ^T	✗ ^{S 2}	✓ ^S
TS	✗ ^{S 1}	✓ ^S	✓ ^S	✓ ^S
Bl	✗ ^{S 2}	✗ ^{S 1}	✓ ^S	✗ ^{S 2}

Table 14: Coffee 4x4 VLA @50 demos — OOD Cup / ID Machine (2/16, 12%).

Machine	Br	WC	Gr	P
K	✗ ^S	✓ ^S	✗ ^{S 5}	✗ ^{S 6}
Bk	✗ ^{T 6}	✗ ^{S 2}	✗ ^{T 2}	✗ ^{S 1}
T	✗ ^{T 2}	✗ ^{S 2}	✗ ^{S 2}	✗ ^{S 2}
W	✗ ^{S 2}	✓ ^S	✗ ^{T 2}	✗ ^{T 1}

Table 15: Coffee 4x4 VLA @50 demos — OOD Cup / OOD Machine (6/12, 50%).

Machine	Br	WC	Gr	P
R	✓ ^{S 7}	✓ ^{S 7}	✗ ^{S 2}	✓ ^S
TS	✗ ^{S 2}	✗ ^{S 7,8}	✗ ^{T 2}	✗ ^{S 1,8}
Bl	✓ ^S	✗ ^{S 2,6}	✓ ^{T 7,9}	✓ ^S

Table 16: Coffee 1x1 VLA @50 demos — ID Cup / ID Machine (1/16, 6%).

Machine	WB	O	Bk	R
K	✓ ^T	✗ ^T	✗ ^S	✗ ^T
Bk	✗ ^{S 1}	✗ ^{T 2}	✗ ^{S 2}	✗ ^{S 2}
T	✗ ^{T 2}	✗ ^{T 2}	✗ ^{S 2}	✗ ^{S 1}
W	✗ ^{T 2}	✗ ^{S 2}	✗ ^{T 2}	✗ ^{T 3}

Table 17: Coffee 1x1 VLA @50 demos — ID Cup / OOD Machine (1/12, 8%).

Machine	WB	O	Bk	R
R	$\times^{T2}$	$\times^{T2}$	$\times^{S2}$	$\times^{T2}$
TS	$\checkmark^T$	$\times^{S2}$	$\times^{S4}$	$\times^{S2}$
Bl	$\times^{T5}$	$\times^{S2}$	$\times^{S4}$	$\times^{T2}$

Table 18: Coffee 1x1 VLA @50 demos — OOD Cup / ID Machine (3/16, 19%).

Machine	Br	WC	Gr	P
K	$\times^{T2}$	$\checkmark^S$	$\times^{T2}$	$\checkmark^S$
Bk	$\times^{T2}$	$\times^{T2}$	$\times^{T2}$	$\times^{S2}$
T	$\times^{S2}$	$\times^{S2}$	$\times^{S1,2}$	$\times^{T10}$
W	$\times^{S1}$	$\checkmark^{T9}$	$\times^{T2}$	$\times^{T1}$

Table 19: Coffee 1x1 VLA @50 demos — OOD Cup / OOD Machine (2/12, 17%).

Machine	Br	WC	Gr	P
R	$\times^{S2}$	$\checkmark^T$	$\times^{T2}$	$\times^{T2}$
TS	$\times^{S2}$	$\times^{S2}$	$\times^{T2}$	$\times^{S5}$
Bl	$\times^{S2}$	$\times^{S6}$	$\times^{S2}$	$\checkmark^T$

Table 20: Pouring 5x3 VLA @105 demos — ID Cup / ID Bowl (10/12, 83%).

Bowl	WB	O	Bk	R	TWC
BL	$\checkmark^S$	$\checkmark^S$	$\checkmark^{T1}$	$\checkmark^{S2}$	$\checkmark^{T3}$
LB	$\checkmark^{S2,3}$	$\checkmark^T$	$\times^T$	$\checkmark$	$\checkmark^{ST3}$
BK	$\checkmark^S$	$\checkmark^T$	$\checkmark$	$\times$	$\checkmark^{T4}$

Table 21: Pouring 5x3 VLA @105 demos — ID Cup / OOD Bowl (7/12, 58%).

Bowl	WB	O	Bk	R	TWC
P	$\checkmark^{S3}$	$\times$	$\checkmark^S$	$\checkmark^S$	$\checkmark^{S6}$
TB	$\times$	$\times^7$	$\times^{11}$	$\checkmark^{T8}$	$\times$
W	$\times^{11}$	$\checkmark^{S9}$	$\checkmark^S$	$\checkmark^{T8}$	$\checkmark^{T2}$

Table 22: Pouring 5x3 VLA @105 demos — OOD Cup / ID Bowl (6/12, 50%).

Bowl	Br	WC	Gr	P
BL	✗	✗ ¹⁰	✗ ¹¹	✓ ^S
LB	✓ ^S	✓ ^{S 2}	✓ ^S	✗ ¹¹
BK	✗ ^{5,6}	✓ ^{T 3}	✓ ^S	✗

Table 23: Pouring 5x3 VLA @105 demos — OOD Cup / OOD Bowl (3/12, 25%).

Bowl	Br	WC	Gr	P
P	✗ ¹¹	✗ ¹⁰	✓	✓ ^{T 2}
TB	✗ ¹¹	✗ ¹¹	✗ ¹¹	✗ ⁸
W	✓	✗ ¹¹	✗ ¹²	✗ ¹²

Table 24: Mug Tree 5x1 VLA @150 demos — ID cups.

Cup	1	2	3	4	5	Rate
WB	✓	✓	✓	✓	✓	100% _{5/5}
O	✓ ¹	✗ ^{1,2}	✓ ¹	✗ ^{1,2}	✓	60% _{3/5}
Bk	✗ ³	✗ ^{3,4}	✗ ^{3,4}	✗ ³	✗ ³	0% _{0/5}
R	✓	✓	✓ ⁴	✗ ³	✓	80% _{4/5}
TWC*	✗ ²	✓	✓ ¹	✗ ²	✗ ³	40% _{2/5}

Table 25: Mug Tree 5x1 VLA @150 demos — OOD cups.

Cup	1	2	3	4	5	Rate
Br	✓	✗ ^{1,3}	✓	✓ ¹	✓ ⁵	80% _{4/5}
WC	✗ ³	✗ ⁶	✓	✗ ⁷	✓	40% _{2/5}
Gr	✓ ¹	✓ ¹	✓ ¹	✗ ²	✗	60% _{3/5}
P	✗ ²	✓	✓	✗	✗	40% _{2/5}
Pu	✓	✓	✓	✗	✗ ²	60% _{3/5}

Table 26: Coffee 4x4 VLA OLD @150 demos — ID Cup / ID Machine (15/16, 94%).

Machine	WB	O	Bk	R
K	✓ ^T	✓ ^S	✗ ^T	✓ ^T
Bk	✓ ^S	✓ ^T	✓ ^S	✓ ^T
T	✓ ^S	✓ ^S	✓ ^S	✓ ^T
W	✓ ^S	✓ ^T	✓ ^T	✓ ^T

Table 27: Coffee 4x4 VLA OLD @150 demos — ID Cup / OOD Machine (7/8, 88%).

Machine	WB	O	Bk	R
R	✓ ^S	✓ ^T	✓ ^T	✓ ^T
TS	✗ ^T	✓ ^S	✓ ^T	✓ ^T
Bl	—	—	—	—

Table 28: Coffee 4x4 VLA OLD @150 demos — OOD Cup / ID Machine (16/16, 100%).

Machine	Br	WC	Gr	P
K	✓ ^S	✓ ^{T O}	✓ ^T	✓ ^S
Bk	✓ ^T	✓ ^S	✓ ^T	✓ ^S
T	✓ ^S	✓ ^S	✓ ^T	✓ ^S
W	✓ ^T	✓ ^T	✓ ^T	✓ ^S

Table 29: Coffee 4x4 VLA OLD @150 demos — OOD Cup / OOD Machine (8/8, 100%).

Machine	Br	WC	Gr	P
R	✓ ^T	✓ ^{S*}	✓ ^T	✓ ^S
TS	✓ ^T	✓ ^{S**}	✓ ^T	✓ ^S
Bl	—	—	—	—